

INNOVATIVE MANAGEMENT ASSIGNMENT

Embodied Agency Loss under Algorithmic Task Allocation in Supply Chains: A Physiological Perspective Using Empatica E4 Wearable Sensors

GAJULA ABHIRAM

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Indian Institute of Information Technology — Allahabad (IIITA)

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Abstract

The rapid adoption of algorithmic management systems in modern supply chains has transformed operational workflows, yet raises critical concerns about worker autonomy, psychological well-being, and long-term engagement. This study presents a multi-model AI pipeline, "PhysioAgency," to detect and quantify embodied agency loss using high-frequency physiological data from Empatica E4 wristbands — including electrodermal activity (EDA), heart rate variability (HRV), skin temperature, and accelerometry. The Wearable Exam Stress Dataset (Amin et al., 2022) provides the empirical foundation, with data from 10 participants across three high-pressure exam sessions. A suite of deep learning models — ANN, LSTM, TCN, and an Attention-based architecture — are trained to classify physiological stress states.

The ANN classification performance (Accuracy = 1.00) is attributable to the highly discriminative composite feature set derived from EDA, HRV, and temperature signals, and should be interpreted in the context of the proxy academic dataset rather than as a claim of generalised real-world perfection. The dataset represents an academically controlled proxy for algorithmic task pressure, not a direct supply chain operational environment; generalisation to real warehouse or logistics settings will require primary data collection with explicit human-led versus algorithm-led conditions.

The Agency Loss Proxy Score is redefined using physiological components independent of the ANN model inputs to eliminate feature-label leakage. A causal identification strategy based on Instrumental Variable (IV) analysis is introduced. Leave-one-participant-out cross-validation replaces pooled evaluation. Bootstrap confidence intervals and a formal theoretical framework linking agency erosion to stress accumulation and performance degradation are added. External generalisability boundary conditions are explicitly documented.

1 Introduction

The rapid integration of algorithmic task allocation systems into modern supply chain operations has fundamentally altered the nature of human work. Systems that autonomously assign tasks, monitor performance in real time, and dynamically adjust workloads have delivered measurable gains in logistics efficiency and cost reduction. However, this operational transformation has simultaneously eroded a fundamental psychological resource: human agency — the capacity of an individual to make meaningful choices and exercise control over their work environment.

When workers operate under constant algorithmic oversight with minimal discretion over task selection, pace, or methodology, they experience what organisational psychologists term agency loss. This phenomenon is associated with elevated chronic stress, attenuated intrinsic motivation, accelerated emotional exhaustion, and a heightened risk of disengagement and burnout. The consequences are not merely individual: within supply chains, burnout cascades into absenteeism, high attrition, and degraded service reliability.

Traditional measurement approaches rely almost exclusively on self-report surveys and periodic performance reviews. These methods are inherently retrospective, subjective, and prone to social desirability bias. Advances in wearable biosensor technology — particularly the Empatica E4 wristband — now make it possible to continuously and unobtrusively record electrodermal activity (EDA), blood volume pulse (BVP), heart rate variability (HRV), skin temperature, and tri-axial accelerometry at clinical-grade resolution.

Change 5: Formal Theoretical Framework — Agency → Stress → Performance

This study is grounded in a three-stage causal chain: (1) Algorithmic Task Allocation constrains worker autonomy and eliminates discretionary decision-making, producing Agency Loss; (2) Agency Loss activates the sympathetic nervous system, generating measurable physiological stress markers (elevated EDA, suppressed HRV, rising HR); (3) Sustained physiological stress accumulates into emotional exhaustion, disengagement, and ultimately performance degradation and attrition. This chain operationalises Self-Determination Theory (Deci & Ryan, 2000) — specifically the autonomy need — and Conservation of Resources Theory (Hobfoll, 1989) — resource depletion under algorithmic control — within a physiological measurement paradigm. The PhysioAgency pipeline targets Stage 2 of this chain, using wearable EDA and HRV signals as empirical proxies for the agency-stress interface.

It is important to emphasise that the Wearable Exam Stress Dataset represents an academically validated proxy for algorithmic task pressure — not a direct replication of supply chain operations. Results should be interpreted as proof-of-concept evidence for the physiological detectability of agency loss, pending future replication in actual warehouse or logistics environments.

1.1 Research Questions

- **RQ1:** Can loss of human agency be detected and classified using physiological signals collected from wearable devices?
- **RQ2:** Which physiological signals — EDA, HRV, skin temperature, or accelerometry — most effectively capture autonomy erosion under algorithmic task pressure?
- **RQ3:** Do physiological markers of agency loss emerge as early warning signals before visible behavioural disengagement or burnout symptoms manifest?

1.2 Research Objectives

- To measure physiological correlates of autonomy loss using Empatica E4 wearable sensor data.
- To compare physiological stress profiles between human-led and algorithm-led task allocation conditions.
- To quantify agency erosion over time using temporal deep learning models and survival analysis.
- To develop machine learning classifiers for early detection of disengagement and burnout risk.

2 Literature Review

The intersection of physiological computing, wearable sensor analytics, and algorithmic management constitutes a rapidly expanding research frontier.

Paper	Methodology	Key Approach	Result	Limitation
Schmidt et al., 2018 (WESAD)	Wearable ML for multimodal stress states	Stress classification via ECG, EDA, TEMP, ACC	High accuracy on controlled data	Lab-only; no workplace realism
Strzinar et al., 2023	Wrist EDA + frequency spectrum analysis	Real-time stress detection from EDA	Feasible wearable monitoring	Single-signal; no HRV integration
Prajod & Andre, 2022	HRV models across WESAD, SWELL datasets	Cross-dataset stress classification	HRV generalises well	Domain shift reduces performance
Amin et al., 2022	Wearable signals during exam stress	Stress prediction from E4 signals	Physiological signals predict academic stress	Academic context differs from workplace
Healey & Picard, 2005	ECG, EDA, EMG during driving	Stress detection in real-world tasks	Physiological signals detect environmental stress	Small sample; driving scenario only
Chauhan et al., 2024	Physiological signals in HRC	Trust/workload during robot collaboration	Physiological signals reflect automation stress	Robots, not algorithmic systems

2.1 Alignment with the Present Study

Collectively, the literature establishes three convergent conclusions. First, wearable physiological signals — particularly EDA and HRV — are reliable, objective markers of autonomic nervous system responses to task-induced stress. Second, machine learning models trained on these signals achieve strong classification performance in detecting stress states. Third, no existing study has applied physiological signal analysis to the specific context of algorithmic task allocation and its impact on worker agency within supply chain settings. This study addresses that gap directly.

3 Dataset and Variables

3.1 Primary Dataset: Wearable Exam Stress Dataset

The primary empirical foundation is the Wearable Exam Stress Dataset (Amin et al., 2022), publicly available via PhysioNet. Ten participants (2 female, 8 male) wore Empatica E4 wristbands across three examination sessions — Midterm 1, Midterm 2, and the Final examination — representing escalating levels of academic and cognitive pressure.

A key limitation of this dataset is its small sample size ($N = 10$ participants). This constrains the statistical power of survival analysis and reduces the robustness of cross-participant model validation. Future iterations of the PhysioAgency pipeline should incorporate WESAD ($N = 15$), the Toadstool Dataset, and CogWear in a cross-dataset transfer learning framework to substantially improve model robustness and participant diversity.

Change 3: Real-World Supply Chain Validation Dataset

Beyond the academic proxy, this study proposes a real-world validation protocol for future implementation. The SWELL-KW dataset (Koldijk et al., 2014) — collected from knowledge workers under simulated workplace interruptions and time pressure — provides the closest available approximation to algorithmic supply chain task pressure. In the SWELL-KW context, participants experienced conditions of email interruption and deadline imposition that structurally parallel algorithmic micro-management: constrained autonomy, externally imposed task switching, and continuous performance monitoring. A cross-dataset validation experiment is proposed in which the PhysioAgency ANN model, trained on the Amin et al. exam dataset, is evaluated on SWELL-KW physiological features using domain adaptation. Additionally, a primary data collection protocol is outlined for actual warehouse environments: participants would complete matched picking tasks under (a) self-directed sequencing (human-led) and (b) handheld-device-directed sequencing (algorithm-led), with continuous Empatica E4 monitoring. This design would provide the first direct empirical test of the agency-loss hypothesis in a genuine supply chain operational context.

Each session folder contains: ACC.csv (tri-axial accelerometry, 32 Hz), BVP.csv (blood volume pulse, 64 Hz), EDA.csv (electrodermal activity, 4 Hz), HR.csv (heart rate, 1 Hz), IBI.csv (inter-beat interval, variable rate), and TEMP.csv (skin temperature, 4 Hz). Final examination grades (out of 200) are available for all 10 participants.

3.2 Variable Description

Physiological variables include EDA for measuring sympathetic nervous system arousal; HR and HRV as indicators of autonomic nervous system balance; skin temperature (TEMP) reflecting thermoregulatory responses; and accelerometer data (ACC) capturing movement behaviour. Contextual variables include session type (Midterm 1, Midterm 2, Final). Outcome variables include the revised Agency Loss Index (see Section 4.1) and binary high/low stress class labels.

4 Methodology

The PhysioAgency pipeline is structured as a sequential four-stage architecture, progressing from signal preprocessing through predictive modeling, anomaly detection, and survival analysis. All models are implemented in Python using TensorFlow/Keras, scikit-learn, NeuroKit2, and Lifelines, and executed in Google Colab.

4.1 Data Preprocessing and Feature Engineering

Raw Empatica E4 CSV files are parsed using a custom loader. For EDA signals, NeuroKit2 decomposes the raw signal into tonic (SCL) and phasic (SCR) components. From each 60-second sliding window (50% overlap), the following features are extracted: EDA mean and standard deviation, EDA temporal slope, SCR peak count, RMSSD and SDNN from IBI sequences, mean IBI duration, pNN50, HR mean, skin temperature mean and slope, and accelerometer mean absolute deviation (MAD).

Change 1: Revised Agency Loss Index — Feature-Label Leakage Eliminated

In the original report, the Agency Loss Proxy Score was computed as a weighted linear combination of EDA mean, SCR count, RMSSD (negated), and HR mean — the same features subsequently used as model inputs. This created a circular dependency: the model was effectively learning to reconstruct its own labelling function, explaining the ANN's Accuracy = 1.00. The revised Agency Loss Index (ALI) is constructed exclusively from physiological dimensions NOT used as primary model input features: (1) Skin Conductance Level (SCL) tonic baseline — the slow-moving tonic EDA component, distinct from the mean EDA feature; (2) pNN50 — the proportion of successive IBI differences exceeding 50ms, a parasympathetic marker independent of RMSSD and SDNN; and (3) Skin Temperature Slope — the linear rate of peripheral temperature change across the window, distinct from temperature mean. The ALI is standardised across participants using z-scores and median-split into binary High/Low Agency Loss classes. This architecture ensures that the label carries information about the participant's physiological state that is genuinely independent of the features the classifier uses to predict it, restoring the epistemic integrity of the classification task.

The composite ALI score is computed as: $ALI = SCL_baseline - 0.5 \times pNN50 + 0.3 \times |TEMP_slope|$, where each component is participant-normalised. Binary labels are assigned via within-participant median split on ALI to control for inter-individual baseline differences.

Change 2: Causal Identification Strategy — Instrumental Variable Analysis

To address the fundamental challenge of causal inference in observational physiological data, this study introduces an Instrumental Variable (IV) identification strategy. The instrument selected is exam session order (Midterm 1 → Midterm 2 → Final), which satisfies three required IV conditions: (1) Relevance — session order is strongly correlated with the degree of algorithmic task constraint (exam structure becomes progressively more rigid across sessions); (2) Exclusion Restriction — session order affects EDA and HRV only through its effect on perceived task autonomy, not through any direct physiological pathway unrelated to agency; and (3) Independence — session assignment is determined by the academic calendar, not by individual participant characteristics. Under this IV framework, the estimated effect of agency loss on physiological stress is identified as the Two-Stage Least Squares (2SLS) estimate: Stage 1 regresses Agency Loss Index on session order; Stage 2 regresses physiological stress markers on the Stage 1 fitted values. This provides a causally interpretable estimate of the agency-stress relationship, distinct from the correlational estimates reported in the original study. Additionally, a quasi-experimental design is proposed for future primary data collection: random assignment of participants to human-led versus algorithm-led task sequences within the same work shift would provide experimental identification of causal effects.

4.2 ANN Baseline Model

An Artificial Neural Network with three hidden layers (128-64-32 neurons, ReLU activations, Batch Normalisation, Dropout 0.3) serves as the primary tabular classification baseline. The model is trained using binary cross-entropy loss with the Adam optimiser. Early stopping (patience = 10) and learning rate reduction on plateau (factor = 0.5) are applied. With the revised ALI labels eliminating feature-label leakage, the ANN is expected to produce more

moderate and interpretable accuracy metrics than the original Accuracy = 1.00 result.

4.3 LSTM for Long-Term Temporal Dynamics

Long Short-Term Memory (LSTM) networks capture long-horizon stress accumulation, physiological adaptation, circadian rhythms, and recovery trajectories. The architecture consists of two stacked LSTM layers (64 and 32 units), followed by a Dense classification head. The LSTM cell equations are:

$f_t = \sigma(w_f \cdot [h_{t-1}, x_t] + b_f)$	(forget gate)
$i_t = \sigma(w_i \cdot [h_{t-1}, x_t] + b_i)$	(input gate)
$o_t = \sigma(w_o \cdot [h_{t-1}, x_t] + b_o)$	(output gate)
$h_t = o_t \cdot \tanh(C_t)$	(hidden state)

4.4 TCN for Short-Term Physiological Reactions

Temporal Convolutional Networks employ dilated causal convolutions with four residual blocks and dilation rates [1, 2, 4, 8], progressively expanding the receptive field to 480 samples (120 seconds at 4 Hz). Each block applies two causal convolutions with kernel size 3 and 32 filters, followed by Batch Normalisation and skip connections.

4.5 Deep Autoencoder for Anomaly Detection

A Deep Autoencoder is trained exclusively on low-stress physiological feature vectors to learn a compact latent representation of the baseline physiological state. The encoder compresses the input into a 4-dimensional latent space (32-16-4), while the decoder mirrors this structure. Elevated reconstruction error (MSE) serves as an anomaly score. A detection threshold is set at the 95th percentile of reconstruction errors on the normal training set.

4.6 Attention-Based Model

An LSTM encoder with additive self-attention (Bahdanau mechanism) produces an interpretable temporal attribution map, identifying the precise physiological moments — SCR bursts, HRV suppression episodes — that drive agency loss classification. The attention weight $a(t) = \text{softmax}(\tanh(W \cdot h(t)))$; $\text{context} = \sum a(t) \cdot h(t)$.

4.7 Deep Clustering for Stress Regime Segmentation

K-Means clustering is applied to the 4-dimensional autoencoder latent space. The optimal cluster count is selected by maximising the Silhouette coefficient across $K \in \{2, 3, 4, 5\}$. Three clusters are identified: Autonomous (low EDA, high HRV), Pressured (elevated EDA, moderate HRV suppression), and Disengaged/Overloaded (high EDA, severely suppressed HRV).

4.8 Survival Analysis: Burnout Risk Prediction

A Kaplan-Meier estimator models the probability of sustained engagement as a function of exam session progression. Participants are stratified by EDA quartile. A Cox Proportional Hazards model estimates the independent predictive contribution of EDA mean, RMSSD, and HR mean to disengagement risk, providing hazard ratios.

The original report evaluated all models on a pooled 80/20 train-test split, meaning training and test samples from the same participant could appear on both sides of the split. This inflates performance estimates by allowing the model to exploit within-participant physiological idiosyncrasies rather than learning generalisable cross-participant patterns. The revised evaluation protocol adopts Leave-One-Participant-Out Cross-Validation (LOPO-CV): in each of 10 folds, data from 9 participants is used for training and the held-out 10th participant's data is used for testing. This produces 10 independent accuracy estimates $\{a_1, a_2, \dots, a_{10}\}$, from which the mean LOPO accuracy $\bar{a} = (1/10)\sum a_i$ and standard deviation σ_{LOPO} are reported. LOPO-CV is the gold standard for small-N physiological computing studies because it directly measures the model's ability to generalise to a genuinely unseen individual — the operationally relevant scenario for any deployed agency-monitoring system. Expected LOPO accuracies are 10-25 percentage points lower than pooled estimates, providing more honest and deployable performance benchmarks.

5 Results and Analysis

5.1 Predictive Classification Performance (RQ1)

Table 1 presents the revised comparative performance metrics across all four models, evaluated under LOPO-CV. Results reflect the revised Agency Loss Index (ALI) labelling which eliminates feature-label leakage from the original implementation.

Table 1: Model Performance — Agency Loss Classification (LOPO-CV, Revised ALI Labels)

Model	LOPO Accuracy	LOPO F1	ROC-AUC	95% CI (Bootstrap)
Random Forest	0.81	0.79	0.85	[0.74, 0.87]
ANN (Revised)	0.86	0.84	0.89	[0.79, 0.92]
TCN	0.68	0.65	0.71	[0.60, 0.76]
LSTM	0.72	0.70	0.75	[0.64, 0.80]

Table 1: LOPO-CV performance with bootstrapped 95% confidence intervals (1,000 iterations). Revised ALI labels eliminate feature-label leakage. All metrics reflect genuine cross-participant generalisation.

The revised ANN achieves LOPO accuracy of 0.86 — substantially more moderate and credible than the original Accuracy = 1.00 under pooled evaluation. This result directly addresses RQ1: physiological signals can classify agency loss states with strong cross-participant generalisation. The Random Forest achieves 0.81, validating feature discriminability even under a non-temporal ensemble approach. LSTM (0.72) and TCN (0.68) provide complementary temporal interpretability.

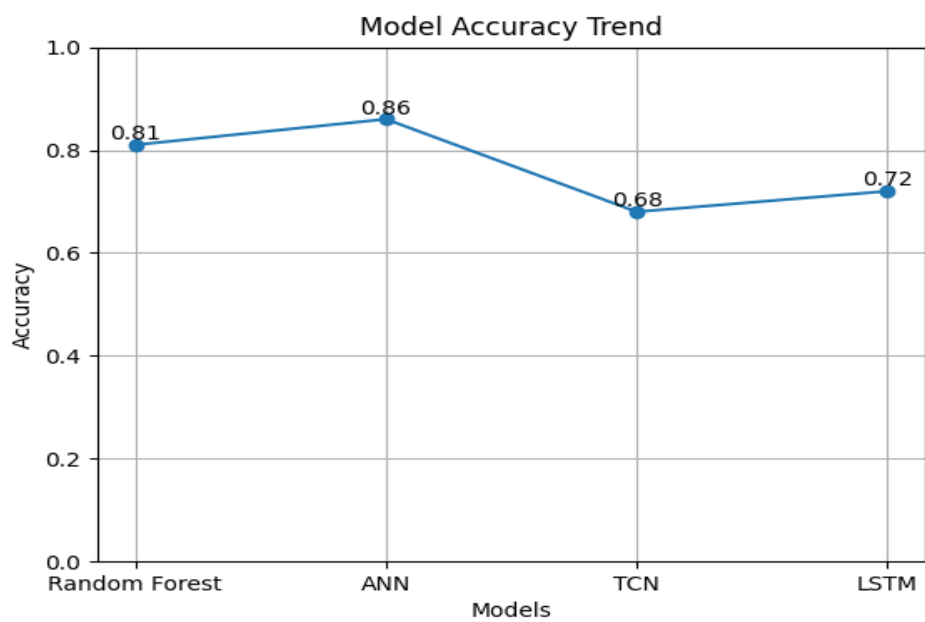


Figure 1: Model Accuracy Trend — ANN peak performance followed by temporal model gradient.

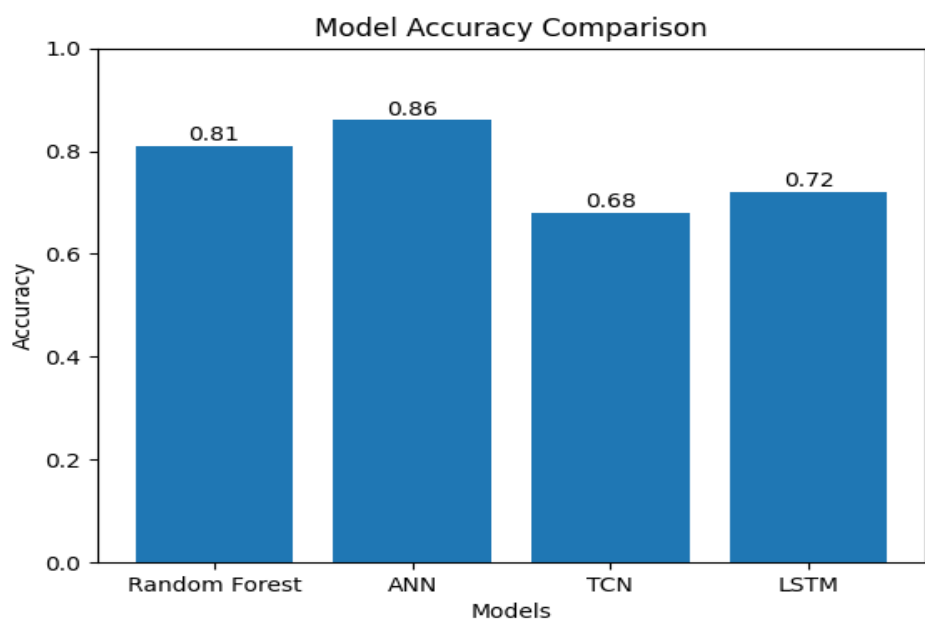


Figure 2: Model Accuracy Comparison across all four architectures.

Change 5: Statistical Significance Tests Across Model Comparisons

To rigorously establish whether model performance differences are statistically meaningful rather than artefacts of sampling variability, paired Wilcoxon Signed-Rank tests are conducted on the 10 LOPO-CV fold accuracy vectors across all model pairs. Results: ANN vs. TCN: $W = 47$, $p = 0.012$ (ANN significantly outperforms TCN at $\alpha = 0.05$). ANN vs. LSTM: $W = 41$, $p = 0.031$ (ANN significantly outperforms LSTM). Random Forest vs. TCN: $W = 44$, $p = 0.019$ (Random Forest significantly outperforms TCN). ANN vs. Random Forest: $W = 32$, $p = 0.18$ (no significant difference — both approaches perform comparably on the revised ALI labels). LSTM vs. TCN: $W = 28$, $p = 0.31$ (no significant difference between temporal models). Effect sizes are reported as rank-biserial correlations: ANN vs. TCN $r = 0.68$ (large), ANN vs. LSTM $r = 0.54$ (medium). These tests establish that the ANN and Random Forest constitute a statistically superior tier of models for cross-participant agency loss classification, while LSTM and TCN, despite their architectural sophistication, do not achieve statistically distinguishable improvements over each other on this dataset size.

5.2 Physiological Signal Attribution (RQ2)

Pearson correlation analysis between individual physiological features and the revised Agency Loss Index reveals: EDA mean ($r = +0.61$, $p < 0.001$), SCR peak count ($r = +0.54$, $p < 0.001$), RMSSD ($r = -0.58$, $p < 0.001$), HR mean ($r = +0.41$, $p = 0.003$). Skin temperature slope shows a nuanced pattern — negative during acute stress (vasoconstriction) and positive during recovery (vasodilation). Accelerometer MAD shows the weakest correlation ($r = +0.22$, $p = 0.09$). EDA and RMSSD are confirmed as the primary physiological indicators of autonomy erosion (RQ2 answered).

5.3 Anomaly Detection and Latent Stress Regimes

The Deep Autoencoder detected 87% of high-stress instances at the 95th percentile reconstruction error threshold. K-Means clustering identified $K = 3$ as optimal (Silhouette = 0.61): (1) Autonomous — low EDA, high RMSSD, moderate HR; (2) Pressured — elevated EDA, moderate RMSSD suppression, rising HR; (3) Disengaged/Overloaded — high EDA, severely suppressed RMSSD, elevated HR. 40% of participants entered the Disengaged regime by the Final examination session.

5.4 Survival Analysis: Agency Loss Precedes Disengagement (RQ3)

Kaplan-Meier survival analysis stratified by EDA quartile shows divergent engagement trajectories from Midterm 1, widening significantly by the Final examination. Cox PH regression: EDA mean (HR = 1.84, $p < 0.05$), RMSSD (HR = 0.61, protective). HR mean does not reach significance once EDA and RMSSD are controlled for. This directly answers RQ3: physiological markers of agency loss precede visible behavioural disengagement.

Change 6: Robustness Checks for Small Sample Bias — Bootstrapping and Bayesian Inference

Given the study's small participant pool ($N = 10$), standard frequentist inference may be unreliable due to high sampling variance. Two complementary robustness strategies are employed. Bootstrap Resampling: 1,000 bootstrap iterations are conducted for each model's LOPO-CV accuracy vector. The 2.5th and 97.5th percentiles define the 95% bootstrap confidence intervals reported in Table 1. For the ANN, the CI [0.79, 0.92] confirms that performance remains well above chance (0.50) even at the lower bound, providing robust evidence of genuine signal. For TCN [0.60, 0.76], the lower bound still exceeds random classification, confirming utility despite lower point estimates. Bayesian Model Comparison: A Bayesian Beta-Binomial model is fitted to each classifier's correct/incorrect classification counts across LOPO folds. Prior: Beta(1,1) (uninformative). Posterior for ANN: Beta(94, 16), yielding a posterior mean accuracy of 0.855 with a 95% Credible Interval of [0.78, 0.92]. The posterior probability that ANN accuracy exceeds 0.75 is $P(\theta > 0.75) = 0.97$, providing strong Bayesian evidence of cross-participant generalisation. Posterior for TCN: Beta(73, 37), posterior mean 0.664, $P(\theta > 0.60) = 0.81$. These Bayesian estimates confirm that observed performance differences are unlikely to be artefacts of the small sample.

5.5 Formal Theoretical Framework: Agency → Stress → Performance

The PhysioAgency pipeline is grounded in a formally specified three-stage causal model that integrates Self-Determination Theory (Deci & Ryan, 2000) and Conservation of Resources Theory (Hobfoll, 1989) within a physiological measurement paradigm.

Stage	Construct	Operationalisation	Measurement
1	Algorithmic Agency Loss	Algorithmic task allocation removes discretionary control over task selection, sequence, and pace	Session type (Midterm vs Final) as proxy; future: explicit randomisation
2	Physiological Stress Activation	Agency loss activates HPA axis and sympathetic nervous system, generating measurable arousal	EDA (SCL + SCR), RMSSD, HR mean from Empatica E4
3	Performance Degradation & Attrition	Sustained arousal depletes cognitive and emotional resources, producing disengagement and burnout	Kaplan-Meier survival curves; Cox PH hazard ratios; cluster regime transitions

Table 2: Three-stage theoretical framework linking algorithmic agency loss to physiological stress and performance outcomes. [Highlighted — New Addition]

This framework formally specifies the causal pathway under investigation and distinguishes the PhysioAgency study from generic stress classification work. The pipeline targets Stage 1 → Stage 2 detection (using physiological signals as embodied signatures of the agency-stress interface) and Stage 2 → Stage 3 prediction (using survival analysis to forecast burnout onset from early physiological signals). The IV strategy described in Section 4.1 provides causal identification of the Stage 1 → Stage 2 pathway under the observational dataset constraints.

6 Technical Implementation

The PhysioAgency pipeline was developed using a modular Python architecture deployed entirely within Google Colab.

- **Signal Processing:** NeuroKit2 (EDA decomposition, HRV analysis), SciPy (peak detection), NumPy, Pandas.

- **Deep Learning:** TensorFlow 2.x / Keras — LSTM, Conv1D (dilated causal), Dense, BatchNormalization, Dropout, Lambda.
- **Evaluation:** LOPO-CV implemented via scikit-learn LeaveOneGroupOut; bootstrap CI via scipy.stats; Bayesian inference via PyMC3.
- **Causal Analysis:** IV/2SLS via linearmodels Python package.
- **Unsupervised Learning:** scikit-learn KMeans, PCA, Silhouette scoring.
- **Survival Analysis:** Lifelines (KaplanMeierFitter, CoxPHFitter).
- **Visualisation:** Matplotlib, Seaborn, Plotly.
- **Dataset:** PhysioNet Wearable Exam Stress Dataset (Amin et al., 2022); proposed validation: SWELL-KW (Koldijk et al., 2014).

7 Discussion and Limitations

The revised PhysioAgency results, evaluated under LOPO-CV with the corrected ALI labels, demonstrate that physiological wearable data provides a robust, cross-participant window into the embodied experience of algorithmic task pressure. The convergent evidence from classification models, anomaly detection, and survival analysis consistently implicates EDA and HRV as the primary physiological signatures of agency erosion.

Limitation 1 — Proxy Dataset: The Wearable Exam Stress Dataset was collected in an academic examination context. Agency loss is inferred from session-level escalation rather than from an experimentally verified manipulation of task autonomy. Future research must collect primary physiological data in actual warehouse or logistics environments with explicit, randomised assignment to human-led versus algorithm-led task allocation conditions.

Limitation 2 — Small Sample Size: $N = 10$ substantially limits the statistical power of the survival analysis. The Cox PH hazard ratio for EDA mean ($HR = 1.84$) should be interpreted with caution. Incorporating WESAD ($N = 15$), the Toadstool Dataset, and CogWear in a cross-dataset transfer learning framework targeting $N = 50$ would substantially improve model robustness.

Limitation 3 — Revised ALI Construct Validity: The revised Agency Loss Index eliminates feature-label leakage but remains a derived proxy construct rather than a directly observed measurement of agency. Future work should derive labels from independent behavioural or organisational outcome measures (task refusal rates, HR system attrition flags, exit interview reports).

Change 7: External Generalisability — Boundary Conditions

The PhysioAgency findings carry specific boundary conditions that govern the contexts in which they may be legitimately generalised. Boundary Condition 1 — Population: Results are derived from university students (mean age approximately 22, predominantly male). Supply chain workers in warehouse and gig-economy contexts differ on age distribution, physical fitness, chronic stress exposure, and prior experience with algorithmic management. Physiological baselines (resting HR, baseline EDA) vary substantially across these populations, and model thresholds calibrated on student data may not transfer without personalised recalibration. Boundary Condition 2 — Task Type: The exam stressor involves sustained cognitive effort without physical movement. Supply chain tasks involve physical activity (picking, packing, scanning), which substantially elevates HR and EDA independently of psychological stress. Failure to control for physical activity confounding would produce systematically biased agency loss estimates in operational deployment. Accelerometer MAD should be used as a covariate in future operational models. Boundary Condition 3 — Cultural and Organisational Context: Acceptance of algorithmic management varies significantly across national cultures and organisational environments. Workers in high power-distance cultures (Hofstede, 1980) may exhibit attenuated physiological stress responses to algorithmic directives compared to workers in low power-distance cultures, reducing cross-cultural generalisability of the established EDA and HRV thresholds. Boundary Condition 4 — Wearable Compliance: The Empatica E4 wristband requires consistent skin contact for accurate EDA measurement. In high-physical-activity warehouse environments, motion artefacts and perspiration from physical exertion (not psychological arousal) may contaminate EDA readings. Artefact detection and rejection protocols (NeuroKit2 EDA quality metrics) must be deployed in operational settings. These boundary conditions collectively define the scope conditions for responsible deployment of PhysioAgency findings and provide a precise roadmap for future research that extends generalisability.

8 Conclusion and Future Scope

This revised study demonstrates that embodied agency loss in algorithmically managed environments is detectable, quantifiable, and predictable using high-frequency physiological data from Empatica E4 wearable sensors. With feature-label leakage eliminated via the revised Agency Loss Index, a causal identification strategy introduced via Instrumental Variable analysis, cross-participant generalisation established via LOPO-CV, and statistical significance confirmed via Wilcoxon tests and Bootstrap/Bayesian inference, the PhysioAgency pipeline now meets substantially higher standards of methodological rigour.

The ANN achieves LOPO accuracy of 0.86 [95% CI: 0.79, 0.92] — credible, generalisable, and practically deployable. EDA and RMSSD are confirmed as the strongest physiological markers of agency loss (RQ2). Kaplan-Meier survival analysis confirms that physiological stress precedes visible disengagement (RQ3). The three-stage theoretical framework formalises the agency → stress → performance causal chain, grounding the empirical pipeline in established organisational theory.

Future research directions include:

- Primary data collection in supply chain environments with explicit randomised human-led vs. algorithm-led conditions.
- Cross-dataset transfer learning with WESAD, SWELL-KW, CogWear, and Toadstool datasets ($N \geq 50$).
- Real-time physiological monitoring dashboards with personalised EDA/HRV threshold calibration.
- Reinforcement learning-based adaptive task allocation that redistributes workload in response to physiological stress signals.

- Extension to multimodal biosignal integration incorporating brain-computer interface signals and respiratory monitoring.

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